THE STARSHIP GALILEO

DEEP SPACE TRAVEL AND COLONIZATION

INSTRUCTIONS MANUAL

THE GEMINI PROTOCOLS

PHASE 04

WHEEL-ARCH PARASITIC KINETIC AIR

Auxiliary generation from the wind the tyres already paid for.
Four arches per vehicle — intake, fan, backward exhaust, tread cooling.
Drag over collection = worthless. Half = good. No scifi.

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Primary Author & Inventor: Archer Quinn, Aerospace Design Engineer, NPhD
Co-Engineers: Gemini 1 AI — Kimi Chat 3 AI

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THE GEMINI PROTOCOLS — INSTRUCTIONS MANUAL — PHASE 04

PHASE 04: WHEEL-ARCH PARASITIC KINETIC AIR — CONTINUOUS AUXILIARY ELECTRICAL GENERATION FOR ELECTRIC VEHICLES, FROM THE WIND THE TYRES ALREADY PAID FOR.

Primary Author & Inventor: Archer Quinn, Aerospace Design Engineer, NPhD — Co-Engineers: Gemini 1 AI, Kimi Chat 3 AI (The Initial Interstellar Co-Engineering Alliance). Project registry: hackaday.io/project/206187. Manual GP-IM-004 v1.0 — 19 August 2026.

STATUS: CURRENT — verdict AERODYNAMIC HYPOTHESIS until the road test measures it (§9). A physical build exists and was photographed on a real vehicle (sitting 60); the geometry was corrected and the drag ruling seated (sittings 61–62). Wording killed 15 August 2026: the word 'siphon' is dead in this phase — the machine is wheel-arch parasitic kinetic AIR. Renamed 12 August 2026 in the Author's naming batches.

§0 — READ THIS FIRST (HOW TO USE THIS MANUAL)

STANDARD NOTATION — MATERIALS & CALIBRATIONS (series law, seated 19 August 2026, sitting 461): Materials or calibrations is not missing science or engineering — it is, to any real physicist or engineer, stating the fkn obvious. Everyone will use different bench test methods, equipment, materials and weight-to-strength choices, where chemical, magnetic or conduction issues are resolved by choice — e.g. carbon fibre over steel. Our designs are base calibrations to existing strength materials: a rim speed does not exceed current basic materials strength. We do not spec methods, and we do not teach welders how to weld.

This manual is written for idiot-proofing — for AIs and for speed. Rules: (1) Build exactly what is written here; every claim is seated law from the master record. (2) Where this phase references another phase, THE DIRECTIONS FROM THAT PHASE ARE INLINED IN §7 — you never leave this document mid-build. (3) Rejected options are listed in §8 so you do not waste time re-deriving them: DO NOT BUILD. (4) The verdict on this machine is empirical — §9 is the only test that settles it; until it runs, the machine stands at AERODYNAMIC HYPOTHESIS. (5) Change control is §10: only the Author's orders change this design.

§1 — WHAT YOU ARE BUILDING

Four auxiliary generators per vehicle — one inside the front wall of each wheel arch. Each unit is a flush-mounted intake scoop with a gravel screen, a sealed inline multi-blade micro-generator fan, and an exhaust duct aimed backwards at the base of the tyre. The fan sits in a wind that ALREADY EXISTS: the spinning tyre blasts air forward and upward into the arch whether the generator is there or not. The machine recovers a fraction of energy the fuel already paid for, from a flow field it may improve. In the

Author's english: there is no drag added to forward motion inside the front of the wheel arch — the wind turbulence is already there; in fact this reduces drag whilst stealing the wind from the tires.

THIS IS AN EARTH PRODUCT — the EV application. The same model was adapted in a different format for the ship: the Phase 4/61 vacuum wheel-arch air principle skims raw nitrogen out of planetary atmospheres and cosmic dust clouds for the Nitropril plant (directions inlined §7.1).

§2 — THE PHYSICS (WHY THERE IS A WIND TO STEAL)

- THE SOURCE: the top of a rolling tyre moves through space at TWICE the vehicle's road speed. The tyre is a centrifugal pump — it shears the air and blasts a chaotic high-pressure jet forward and upward into the front section of the wheel well. That pocket is why wheel arches are notorious hubs of turbulence and pressure build-up — it is EXISTING drag, pushing against the car right now.
- THE SPENT-PETROL DOCTRINE (the Author, verbatim): "i dont thinks its free, its spent petrol, hot tyres are free heat from spent petrol... the wind is not drag, it was already paid for in petrol". The machine claims NO free energy: the energy spinning that air comes from the engine's torque on the tyre against the road. It is parasitic kinetic waste. Recovering waste is not magic; it is accounting.
- THE PRESSURE-RELIEF ARCHITECTURE (the Author, verbatim): "the front of the wheel arch air automatically goes to ground, off the tyre hits the front wall and pressure from the continuous air pushes it down, the fan is merely between the wall and ground... the fan is about 1/3 the size of that wall space, so it isnt a pressure choke point, the full arch is still open, a larger area cannot create an increase in pressure as path of least resistance has increased... the previous pressure area is now lessor by 1/3 backpressure. I will conceede any point if you can show me where a drag occurs".
- THE CHAIN: engine turns wheel -> wheel violently accelerates air inside the arch -> capture vent isolates the high-velocity jet -> fan blade spins -> generator makes electricity -> net system efficiency goes UP, because you reclaimed from a pre-existing pocket of chaos instead of sticking a windmill on the roof.

§3 — THE INTAKE MATRIX

- LOCATION: the front interior quadrant of ALL FOUR wheel wells. The target air mass is the centrifugally accelerated forward/upward jet off the tyre shear rotation, moving at 2x vehicle speed.
- INTERFACE: a structural intake scoop FLUSH-MOUNTED into the forward wheel-well liner — intercepting the pre-existing high-pressure parasitic pocket. Flush means flush: nothing proud into clean air. A proud scoop is the rejected option (§8).
- FILTRATION: a high-durability STAINLESS STEEL MESH gravel screen over the intake — coarse gravel, pebbles and heavy debris are physically deflected and drop back to the road; the high-velocity air

passes through to the fan. The arch is an industrial sandblaster at highway speed; build the screen like it.

§4 — THE GENERATION CORE

- TURBINE: an internal multi-blade micro-generator fan assembly, INLINE in the duct. The fan occupies about 1/3 of the wall space — the arch stays open around it; the unit is a relief valve, not a plug.
- HOUSING: a sealed inline conduit insulated against moisture intrusion, road spray and fine dust particulates.
- WEAR: internal channels optimized for micro-abrasion (sand) WITHOUT compromising rotor balance. Balance is life: an abraded, unbalanced rotor in a wheel well is a failure you designed in.

§5 — THE EXHAUST & COOLING MATRIX

- VECTOR: exhaust directed strictly BACKWARDS toward the rear base of the tyre. Never sideways (§8).
- THE COOLING LOOP: the exhaust stream is a continuous forced-convection cooling jet over the tyre base tread — tyres absorb friction heat, carcass flex heat and brake-rotor radiant heat; excessive heat kills rubber and spikes pressure. The unit's waste air is the tyre's cooling system. No separate fan required.
- THE SCAVENGE: the zone immediately behind a rolling tyre is a turbulent LOW-PRESSURE wake. Venting into it lets the atmosphere PULL the air through the duct — a natural scavenging effect that eliminates internal backpressure and stops turbine stall. Exhaust into a high-pressure zone and the duct backs up, the blades stall, and the generator chokes.

§6 — POWER MANAGEMENT & BUILD SEQUENCE

The electrical side is a multi-channel wide-input hybrid charge regulator feeding directly into the high-voltage main EV battery bus. Four generators, four channels, one bus.

BUILD SEQUENCE:

- 1. SCREEN AND SCOOP: cut the flush intake into the forward wheel-well liner; fit the stainless gravel screen. Nothing proud of the liner surface.
- 2. CORE: mount the sealed inline fan-generator in the duct, fan spanning about 1/3 of the wall space, between the front wall and the ground path the arch air already takes.
- 3. DUCT: run the exhaust backwards to the rear base of the tyre, exiting into the low-pressure wake; aim the stream across the tread base for the cooling effect.

- 4. REPEAT x4: all four arches, same build.
- 5. WIRE: one regulator channel per unit; wide-input hybrid charge regulator; output to the HV main battery bus.
- 6. TEST: the road protocol of §9 — the ONLY test that settles this machine.

§7 — CROSS-PHASE DIRECTIONS, INLINED (PROGRAM LAW: DIRECTIONS, NOT POINTERS)

§7.1 To PHASE 128 via the PHASE 4/61 VACUUM WHEEL-ARCH AIR — THE SHIP-SIDE ADAPTATION (what the same model does up there)

- The adapted model: the Phase 4/61 vacuum wheel-arch air principle SKIMS raw, free nitrogen gas out of ambient planetary atmospheres or cosmic dust clouds. The wheel's trapped turbulence down here is the skimmed flow up there — same harvest logic, different medium.
- What it feeds (Phase 128, seated law): an industrial-scale Haber-Ostwald Ammonium Nitrate PRILLING PLANT on the sub-deck floor converts the raw gas into stable, dry NITROPRIL rock-gravel on demand. The reserve rides as cold, un-pressurized, solid-state crystal — it cannot leak, boil, or explode; heavy liquid cooling tanks are eliminated. (Nitropril housing detail: Phase 88 — solid-state, un-pressurized granules, never a pressurized gas block.)

§8 — REJECTED AND SUPERSEDED OPTIONS (DO NOT BUILD)

- REJECTED — the bodywork windmill: a scoop or turbine in the CLEAN outer airflow is a parachute; it adds frontal drag and burns more than it recovers. The whole point is that the arch wind is NOT clean air — it is trapped, detached, already-paid turbulence.
- REJECTED — side venting: exhaust through the fender sandblasts the paint with grit over time and splashes mud and wet grime over doors and mirrors in rain. The Author ruled it out on the build photographs; backwards at the tyre base is the law.
- REJECTED — front-of-vehicle mounting, OUTSIDE the arch (the Author's drag ruling, sitting 62, verbatim): "no we will skip that one, that would cause drag not free energy, it must be in front of the tyre inside an arch, that wind is already there". An earlier photographed build sat outside the arch; the ruling moved it inside. Inside, ahead of the tyre. Nowhere else.
- SUPERSEDED — the laboratory protocol (sitting 148's three-configuration lab test): killed by the Author's test ruling (sitting 149, verbatim): "you cannot do it in a lab there is no venturi, no low pressure behind the car, there is no thermal updraught from the road, every car is a different size every car weight drops as fuel goes down." The physics the lab would need to hold constant does not exist off the road.

§9 — OWED TO BENCH (THE ONLY TEST, AND THE KILL/KEEP THRESHOLD)

THE ROAD TEST (the Author's ruling — the only possible test): in battery cars, connect the device to a SEPARATE set of batteries and measure output and run distance of cars WITH and WITHOUT, carrying the weight of the extra battery being charged. Same route, same speeds, honest weather.

THE KILL/KEEP THRESHOLD IS LAW (the Author, verbatim): "if there is a drag over collection its worthless, if its half im good, no scifi or wishful thinking". Measure, don't argue (Draw Rule). Until the road numbers exist, this machine stands at AERODYNAMIC HYPOTHESIS — no drag penalty established by argument, net effect unmeasured. OWED FIGURES: watts per unit at highway speed; net drag delta with and without; charge recovered per 100 km against the extra battery mass carried.

§10 — STATUS, PROVENANCE, CHANGE CONTROL

Provenance: Phase 4 body seated in the vetting era — vetting batch 4 (sitting 148: flow-path clarification + kill/keep threshold), batch 6 (11 August 2026: pressure-relief architecture + spent-petrol doctrine, verbatim); the Author's test ruling (sitting 149) killed the lab protocol. The render of record was accepted by the Author — "close enough" (sitting 25); the pencil sketch governs (sitting 59, Draw Rule: pencil outranks render); the physical build was photographed on a real vehicle (sitting 60: real hardware outranks pencil and render), geometry-corrected (sitting 61) and drag-ruled (sitting 62). Merged 14 August 2026 (duplicate texts consolidated); wording killed 15 August 2026 ('siphon' is dead). Standard notation per sitting 461. Every claim in this manual is seated law from the master record, v2.1.

CHANGE CONTROL: this design changes only on the Author's orders, seated as sittings in the master record. If the master and this manual ever disagree, THE MASTER WINS — and the manual is re-issued with the amendment inlined. Manual version: v1.0, 19 August 2026 — first issue (fourth manual of the Instructions Manual program: 'idiot proofing for AIs and speed... do them one at a time').

END OF MANUAL — PHASE 04.